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CONTENTS

1	Advances and Trends in Micro- and Nano-Fabrication Rashad Abaszade, Elmira Khanmamadova	1-23
2	Application of drones to hybrid energy systems: operating cost reduction and advantages N.T.Vahabli	24-28
3	Economic and environmental studies in the sustainable development of renewable energy in Karabakh M.Huseynova	29-32
4	Comparative analysis of oil-contaminated soil remediation methods A.Azizov, S.Huseynli	33-35
5	Development of an Automated Control System for Crude Oil Heater and Its U.Rzayev, İ.Bayramov	36-38
6	Electromagnetic waves and their effects on the human organism T. Huseynov, B. Davudov	39-42
7	Investigating the impact of climate change on the caspian sea level and ecological conditions A.Khalilova, N.Farzaliyeva	43-48
8	Laser strengthening of CP65+5%MF abrasive material for gears of oil pumps A.Guliyev, Y.Jahangirova	49-56
9	Plastic Alternatives: Bioplastics and Nanotechnology L.H. Memmedova, F.F. Ismayilova	57-59
10	Rehabilitation, modernization and expansion of electrical networks in liberated territories K.Gulmammadov	60-68
11	Renewable energy sources in the territory of east zangezur and Karabakh S.Gasimova, G.Mammadova	69-71
12	Sustainable urbanization and green infrastructure for reducing carbon emissions S.Gasimova, N.Əhədzadə	72-75
13	Solutions for optimizing heat dissipation on the printed circuit board of the drive circuit of a DC motor Jamil Gunashov, Rashad Abaszade	76-82
14	Prompt-based forecasting of monthly solar irradiation in the fuzuli district using ai models Mobil Maharramov	83-86
15	Industrial waste processing and utilization: a study of Sumgayit technopark C.A.Gurbanov	87-90
16	Artificial Intelligence and Geology F.M.Hajizadeh, N.Y.Ahadi	91-95
17	Energy Sector Vulnerability Fidan Abdullazade, Farhad Aliyev	96-99
18	Water Degradation Challenges in Azerbaijan's Karabakh Region S. R. Yusifova	100-103
19	Waste collection and utilization at sumgait technologies park C.A.Gurbanov	104-106

Application of drones to hybrid energy systems: operating cost reduction and advantages

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Abstract: The integration of unmanned aerial vehicles (UAVs) with hybrid energy systems is rapidly emerging as a revolutionary approach to managing and optimizing energy generation, distribution, and maintenance. This article explores the significant potential for reducing operating costs through the use of drones in hybrid energy systems, as well as additional benefits that go beyond mere cost savings. The article covers the current state of hybrid energy systems, the technical evolution of drones, their synergies, and the economic, environmental, and strategic advantages they collectively offer.

Keywords: drones, hybrid energy systems, cost reduction, predictive maintenance, autonomous operations, sustainability

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1. Introduction

Hybrid energy systems that combine various renewable sources such as solar, wind and storage systems are becoming increasingly vital in the transition to sustainable energy. The complexity of managing such systems requires innovative solutions to minimize maintenance costs, increase efficiency and optimize overall performance. Drones provide a transformative tool for operators managing these diverse energy assets with the ability to perform high-resolution inspections, data acquisition and real-time monitoring.

In recent years, the adoption of drones in the industrial and energy sectors has accelerated due to advances in sensor technology, machine learning and communications. The application of drones in hybrid energy systems is not just an incremental improvement, but also a paradigm shift, especially in terms of reducing operating costs. By deploying drones for regular inspections, maintenance assessments and system monitoring, operators can significantly reduce costs associated with manual labor, safety risks and unplanned downtime.

This article examines the technical and economic aspects of integrating drones into hybrid energy systems. It explores technical architecture, operational challenges, and measurable cost reductions. The discussion also includes broader strategic benefits such as improved data analytics, predictive maintenance, and improved environmental monitoring capabilities.

Hybrid power systems combine multiple power generation technologies, such as solar photovoltaics, wind turbines, and energy storage solutions, to provide a stable, efficient, and sustainable power supply. Their inherent diversity allows them to mitigate the intermittency of renewables and balance load fluctuations.

However, the integration and management of these systems is inherently complex and involves several components:

- Power generation units: Solar panels, wind turbines, and other renewables.
- Energy storage: Batteries or other forms of energy storage that help balance supply and demand.
- Control systems: Advanced controllers that optimize power flow, ensure stability, and integrate predictive models.
- Distribution grids: The interconnected network infrastructure that distributes generated energy to end users.

The multifaceted nature of these systems requires continuous monitoring and maintenance. Historically, this has been achieved through manual inspections and scheduled maintenance, which, while effective, are time-consuming and prone to human error. This is where drone technology offers a significant advantage [1-17].

2. Experimental detail

One of the main advantages of using drones is their high mobility. Drones monitor large areas of power plants, solar panels and wind turbines with high accuracy, detecting faults and potential risks in time. Such real-time information allows for operational control of the system and preventive maintenance measures. While inspections take hours in traditional methods, drones perform this process accurately and quickly, providing significant savings in operating costs. Also, the role of drones in the operating process is not limited to monitoring. They also enable analysis of energy efficiency in the system through the high-quality images and thermal data they collect. Based on this data, it is

determined which parts of the system are facing the most problems and at which points energy loss occurs. Thus, the application of drones also becomes a valuable source of information for the effective implementation of energy production optimization and distribution.

Regarding the shortcomings of traditional monitoring methods, it is observed that inspections based on human labor are expensive in terms of time and money. Inspections, especially in large areas, require a large portion of human resources, and as a result, operating costs increase dramatically. The fact that drones automate this process minimizes resource waste and creates opportunities for rapid intervention. For example, data collected by drones through real-time monitoring allows for the detection of malfunctions at an early stage, thus preventing major errors and accidents. As a result, significant savings are achieved in operating costs.

The application of this technological solution is not only limited to reducing operational maintenance costs but also creates conditions for increasing the overall performance and reliability of the system. This is also of great importance in terms of ensuring worker safety. In particular, the application of drones in mountainous, forested and other inaccessible areas ensures both uninterrupted continuation of operations and minimizes potential safety risks.

Also, the analysis of data collected by drones plays an important role in providing additional advantages necessary for the optimization of energy systems. For example, the performance of solar panels, the rotation of wind turbines and other technical parameters are analyzed using data collected by drones. As a result of this analysis, it is determined which parts require maintenance, where energy is lost and what measures should be taken to optimize the overall system. As a result, the system operates more efficiently in the long term and operating costs are reduced.

The main advantage of integrating drones into hybrid energy systems is a significant reduction in operating costs. Traditional maintenance strategies often involve high labor costs, long downtimes, and the risks associated with manual inspections. In contrast, drone operations can be automated or remotely controlled, offering several cost-effective advantages.

A study conducted by the United States National Renewable Energy Laboratory (NREL) found that labor accounts for 50-70% of operating costs in renewable energy plants. In addition, unexpected failures due to delayed inspections can lead to millions in lost revenue in downtime.

The formula for Operational Cost (OC) without drones is:

$$OC_{manual} = (L \times R_L) + (D \times C_D) + (E \times C_E)$$

L= Labor hours

R_L= Labor rate per hour

D= Downtime hours

C_D= Cost of downtime per hour

E= Equipment rental costs

The formula for drone-based cost savings:

$$OC_{drone} = (F \times R_D) + (M \times C_M)$$

F= Flight time (hours)

R_D= Drone operator rate (typically lower than technician rates)

M= Maintenance cost of drone equipment

The cost comparison includes both direct savings (such as reduced labor costs) and indirect savings (such as improved system uptime and early detection of failures). Studies show that integrating drone technology can reduce operating costs by 30-50% over the life cycle of a hybrid energy system.

A simplified formula for estimating operating cost reductions can be expressed as follows:

$$Cost\ Reduction(\%) = \frac{C_{traditional} - C_{drone}}{C_{traditional}} \times 100$$

C_{traditional}= the average cost of maintenance and inspection using traditional methods.

C_{drone}= the cost when using drones, including initial investment, operational expenses, and periodic maintenance of the drone fleet.

In addition to cost savings, the application of drones to hybrid energy systems offers several additional benefits that enhance operational efficiency and strategic planning.

One of the main advantages of using drones in power system management is improved safety. Traditionally, technicians have been required to work on top of tall towers, near high-voltage equipment, or in hazardous conditions such as adverse weather conditions. Using unmanned aerial vehicles, operators can remotely inspect and monitor these facilities, thereby reducing the risk of accidents and occupational injuries. This not only protects human resources but also minimizes potential liability and insurance costs.

Drones are capable of collecting high-quality images and sensor data with exceptional accuracy. This accuracy is crucial for detecting early signs of wear, damage, or other anomalies that may indicate future failures. Integrating real-

time data collection systems means operators receive timely warnings, allowing them to resolve problems before they become costly failures. The ability to continuously monitor, automate, and monitor represents a significant improvement over periodic manual inspection.

By harnessing the power of drones, energy operators can plan maintenance more effectively. Instead of relying on routine inspections that may miss intermittent or emerging issues, drones can be programmed to conduct frequent surveys, ensuring that maintenance interventions are both timely and accurate. This proactive approach to maintenance reduces unplanned outages and keeps the power system operating at peak efficiency. The result is a more robust and reliable power supply that is less prone to outages.

The rich data set generated from drone operations can be integrated with advanced predictive analytics. Machine learning algorithms can analyse trends, predict potential failures, and optimize maintenance schedules. The following formula encapsulates a predictive maintenance model:

$$MTBF = \frac{T_{operational}}{N_{failures}}$$

MTBF= Mean Time Between Failures.

T_{operational}= total operational time.

N_{failures}= number of failures observed over that time period.

By increasing MTBF through proactive interventions enabled by drone data, operators can further reduce the frequency and severity of system downtime, thus protecting revenue and increasing system reliability.

Integrating drones into a hybrid energy system requires a robust communications network and data management platform. The overall architecture typically includes the following layers:

- Data Acquisition Layer: This is where drones collect imaging, thermal, LiDAR, and sensor data. The quality of data acquisition is paramount, and this requires high-quality sensors and reliable transmission capabilities.
- Data Processing Layer: Once data is captured, it is transmitted to centralized servers or cloud-based platforms, where it is processed using image recognition algorithms, machine learning models, and statistical analysis tools. This layer is critical for transforming raw data into actionable insights.
- Data Analytics and Visualization Layer: The processed data is then visualized in the form of charts, graphs, and detailed reports. Operators can monitor system health in real time and review historical data to identify trends.
- Control and Audit Layer: Finally, insights gained from data analytics inform control systems and maintenance planning. Automated alerts and predictive models facilitate timely interventions, optimizing the overall performance of the hybrid energy system.

For effective integration, drone systems are often connected to existing Internet of Things (IoT) networks and SCADA systems. This connectivity allows for real-time monitoring and seamless communication between the drones and the central control unit. The following formula can be used to model the communication efficiency between these components.

$$\eta = \frac{D_{successful}}{D_{total}} \times 100$$

η= communication efficiency as a percentage.

D_{successful}= number of data packets successfully transmitted.

D_{total}= total number of data packets sent.

Given the critical role drones play in monitoring and maintenance, it is important to incorporate robust security protocols and precautions. These can include redundant communication channels, dual sensor arrays, and fail-safe mechanisms that allow drones to return to a safe landing zone in the event of an emergency. Incorporating these measures not only protects the drones themselves but also ensures the continuity of data collection and system monitoring.

Looking ahead, the future of drone integration in hybrid energy systems is incredibly promising. One of the most exciting developments is the emergence of fully autonomous drones powered by advanced artificial intelligence. Soon, these drones are expected to operate without the need for constant human intervention, making decisions in real time through sophisticated machine learning algorithms. They will autonomously detect faults, prioritize maintenance tasks, and dynamically adjust flight paths and inspection frequencies based on evolving conditions of the energy infrastructure. This will not only reduce response times and operating costs, but also significantly increase the reliability and overall performance of hybrid energy systems.

At the same time, continued advances in sensor technology promise to elevate the capabilities of drone inspections to unprecedented levels. Future drones are expected to be equipped with next-generation sensors such as hyperspectral imaging, state-of-the-art LiDAR, and even emerging quantum sensors, all of which will allow them to detect microscopic anomalies that are currently beyond the reach of conventional equipment. The high-resolution data

obtained will allow energy operators to fine-tune their predictive maintenance models, thus preventing failures before they occur and reducing the risk of system downtime. As the quality and volume of data increases, integrating these insights with advanced big data analytics and cloud computing platforms will allow decision makers to optimize energy generation and system performance more effectively than ever before.

In addition, the integration of drone technology with renewable energy forecasting and smart grid management systems will revolutionize energy generation and distribution. By incorporating real-time meteorological data and energy consumption patterns, drones will provide instant feedback to control systems, enabling adaptive adjustments to energy production and storage. This real-time interaction ensures that hybrid energy systems can operate at peak efficiency, even when environmental conditions are fluctuating. Drones will thus play a key role in stabilizing energy production and supporting a dynamic balance between energy supply and demand.

Another promising development lies in the potential of collaborative multi-drone systems. Unlike the traditional model where drones operate in isolation, future systems will feature networked drones that communicate and coordinate their activities in real time. This collaborative approach not only ensures comprehensive coverage of large and complex infrastructures but also introduces redundancy to the system; if one drone experiences a technical problem, others can quickly fill the gap, thereby maintaining continuous and effective monitoring. As hybrid energy systems expand, such network operations will be particularly valuable, offering scalable solutions that can adapt to increasing operational requirements without a corresponding increase in costs.

Globally, trends in environmental policy and an increasing focus on sustainability will further accelerate the adoption of drone technology in the energy sector. Governments and regulators around the world are beginning to recognize the multifaceted benefits of integrating drones into hybrid energy systems, from enhanced safety and reduced environmental impacts to significant cost savings and operational efficiencies. Future policy frameworks are expected to include incentives such as tax breaks, grants, and simplified regulatory processes, all designed to reduce the financial barriers to drone integration. These measures will not only encourage early adoption but also foster an environment of rapid technological innovation and widespread adoption of drone-assisted energy management.

8. Conclusion

The integration of drone technology into hybrid energy systems is a transformative step forward in energy management. The significant reduction in operating costs through improved safety, real-time data collection, and predictive maintenance, combined with additional benefits such as improved environmental monitoring and scalability, create a compelling case for widespread adoption of this technology. The detailed analysis presented in this article highlights the potential of drones to reshape how hybrid energy systems are maintained and optimized. While challenges such as regulatory compliance, technological limitations, and cybersecurity risks remain, ongoing innovation and strategic investments are expected to overcome these barriers. The prospects for autonomous, AI-driven drones, advanced sensor technologies, and collaborative multi-drone systems are particularly promising, and the benefits of this approach are only likely to increase over time. As a result, the introduction of drones into hybrid energy systems not only offers significant cost savings, but also improves overall operational efficiency, safety, and sustainability. As the energy sector evolves, operators that leverage these innovative technologies will be well positioned to achieve optimal performance, ensuring a more sustainable and environmentally responsible energy supply for the future.

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A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a cloudy sky. The image is framed by a dark green curved border at the top and bottom.

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A graphic of a water splash or wave, rendered in a soft, painterly style, located in the bottom right corner of the page.